

4.6 Variation of Parameters

The method of undetermined coefficients (section 4.4) can be used to find a particular solution of $ay'' + by' + cy = g(x)$

When: ① a, b, c are constants

② $g(x)$ is a sum of
[Polynomial] $\times$ [Exponential] $\times$ [sinusoid].

In this section we will present a method to find a particular solution of any nonhomogeneous linear DE of order two.

The method was developed by the famous astronomer and mathematician Lagrange.

We present the method of variation of parameters for DEs of order two but it can be extended to n^{th} -order DEs.

Consider the DE :

$$y'' + P(x)y' + Q(x)y = f(x) \quad [NH]$$

in standard form .

Goal : We want to find a particular solution y_p of $[NH]$.

Assume that we know the homogeneous solution :

$$y_c = c_1 y_1 + c_2 y_2$$

Note : y_c is the general solution of

$$y'' + P(x)y' + Q(x)y = 0 \quad [H]$$

We look for y_p in the form :

$$y_p = u_1 y_1 + u_2 y_2$$

for two functions $u_1 = u_1(x)$ and $u_2 = u_2(x)$.

We assume that :

$$\textcircled{1} \quad u_1' y_1 + u_2' y_2 = 0$$

and

$$\textcircled{2} \quad y_p \text{ is a solution of } [NH].$$

To find u_1 and u_2 we plug y_p in the DE $[NH]$.

$$y_p = u_1 y_1 + u_2 y_2$$

$$\Rightarrow y'_p = \underline{u'_1 y_1} + u_1 y'_1 + \underline{u'_2 y_2} + u_2 y'_2$$

from assumption ①, $u'_1 y_1 + u'_2 y_2 = 0$.

$$\Rightarrow y'_p = u_1 y'_1 + u_2 y'_2$$

$$\Rightarrow y''_p = u'_1 y'_1 + u_1 y''_1 + u'_2 y'_2 + u_2 y''_2$$

Plug y_p , y'_p , and y''_p into

[NH] to get:

$$y_p'' + P(x)y_p' + Q(x)y_p = f(x)$$

$$\left[\underbrace{u_1' y_1'} + \underbrace{u_1 y_1''} + u_2' y_2' + \underline{\underline{u_2 y_2''}} \right] + P(x) \left[\underbrace{u_1 y_1'} + \underline{\underline{u_2 y_2'}} \right]$$

$$\dots + Q(x) \left[\underbrace{u_1 y_1} + \underline{\underline{u_2 y_2}} \right] = f(x)$$

Collect terms to get:

$$u_1 \left[y_1'' + P(x)y_1' + Q(x)y_1 \right] + u_2 \left[y_2'' + P(x)y_2' + Q(x)y_2 \right]$$

$$\dots + u_1' y_1' + u_2' y_2' = f(x)$$

Since y_1 and y_2 are homogeneous solutions, then:

$$\begin{cases} y_1'' + P(x)y_1' + Q(x)y_1 = 0 \\ y_2'' + P(x)y_2' + Q(x)y_2 = 0 \end{cases}$$

$$\Rightarrow u_1' y_1' + u_2' y_2' = f(x)$$

Combining this with assumption ①
we obtain :

$$\begin{cases} u_1' y_1 + u_2' y_2 = 0 \\ u_1' y_1' + u_2' y_2' = f(x) \end{cases}$$

This is a linear system for u_1' and u_2' . We can write it in matrix form as :

$$\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ f(x) \end{bmatrix}$$

We can solve the system by using Cramer's rule to get:

$$u_1' = \frac{W_1}{W} \quad \text{and} \quad u_2' = \frac{W_2}{W}$$

where:

$$\begin{cases} W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \quad \leftarrow 2 \times 2 \text{ determinant} \\ W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix} \\ W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix} \end{cases}$$

Note: W is the Wronskian of y_1, y_2 .

Since y_1, y_2 are LI, then $W \neq 0$.

$$\begin{cases} u_1 = \int \frac{w_1}{w} dx \\ u_2 = \int \frac{w_2}{w} dx \end{cases}$$

Finally, $\boxed{y_p = u_1 y_2 + u_2 y_1}$

Note: We don't need constants of integration when we find u_1 and u_2 .

The general solution of $[NH]$ is

$$y = y_c + y_p$$

$$y = c_1 y_1 + c_2 y_2 + y_p$$

Any constant of integration for u_1 or u_2 could be absorbed in c_1 or c_2 .

ex: ① Solve $y'' - 4y' + 4y = 3e^{2x}$

Step ① Find y_c

The DE has constant coefficients.

Roots : $m^2 - 4m + 4 = 0$

$$(m - 2)^2 = 0$$

$$m = 2$$

$$\Rightarrow y_c = c_1 e^{2x} + c_2 x e^{2x}$$

Step ② Find y_p

We will use the method of Variation of parameters. We need :

$$\begin{cases} y_1 = e^{2x} \\ y_2 = x e^{2x} \end{cases} \rightarrow \text{from } y_c$$

$$f(x) = 3e^{2x} \leftarrow \text{from the DE in standard form.}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & e^{2x} + 2xe^{2x} \end{vmatrix} = e^{4x}$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix} = \begin{vmatrix} 0 & xe^{2x} \\ 3e^{2x} & e^{2x} + 2xe^{2x} \end{vmatrix} = -3xe^{4x}$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix} = \begin{vmatrix} e^{2x} & 0 \\ 2e^{2x} & 3e^{2x} \end{vmatrix} = 3e^{4x}$$

$$u_1 = \int \frac{W_1}{W} dx = \int -3x dx = -\frac{3}{2} x^2$$

$$u_2 = \int \frac{W_2}{W} dx = \int 3 dx = 3x$$

$$y_p = u_1 y_1 + u_2 y_2$$

$$y_p = \left(-\frac{3}{2} x^2\right) e^{2x} + 3x (xe^{2x})$$

$$y_p = \frac{3}{2} x^2 e^{2x}$$

The general solution is

$$y = y_c + y_p$$

$$y = c_1 e^{2x} + c_2 x e^{2x} + \frac{3}{2} x^2 e^{2x}$$

Note: If we use the method of undetermined coefficients (4.4) to find y_p , we would look for

$$y_p \text{ as } y_p = A x^2 e^{2x}$$

Since we have a bad case. This is consistent with the $y_p = \frac{3}{2} x^2 e^{2x}$ that we found.

ex: ② Solve $2y'' + 18y = \sec(3x)$

Step ① Find y_c

Roots: $2m^2 + 18 = 0$

$$m^2 = -9 \rightarrow m = \pm 3i$$

$$y_c = c_1 \cos(3x) + c_2 \sin(3x)$$

Step ② Find y_p

Variation of Parameters needs:

$$\left\{ \begin{array}{l} y_1 = \cos(3x) \\ y_2 = \sin(3x) \end{array} \right\} \text{ from } y_c$$

$$f(x) = \frac{1}{2} \sec(3x)$$

← from the standard form.

⚠ Don't forget that $f(x)$ is from the

standard form: $y'' + 9y = \underbrace{\frac{1}{2} \sec(3x)}_{f(x)}$

$$W = \begin{vmatrix} \cos(3x) & \sin(3x) \\ -3\sin(3x) & 3\cos(3x) \end{vmatrix} = 3$$

$$W_1 = \begin{vmatrix} 0 & \sin(3x) \\ \frac{1}{2} \sec(3x) & 3\cos(3x) \end{vmatrix} = -\frac{1}{2} \tan(3x)$$

$$W_2 = \begin{vmatrix} \cos(3x) & 0 \\ -3\sin(3x) & \frac{1}{2} \sec(3x) \end{vmatrix} = \frac{1}{2}$$

$$u_1 = \int \frac{w_1}{w} dx = \int \frac{-\tan(3x)}{6} dx$$

$$u_1 = -\frac{1}{6} \int \frac{\sin(3x)}{\cos(3x)} dx$$

$$\begin{aligned} \text{let } u = \cos(3x) &\rightarrow du = -3 \sin(3x) dx \\ &\rightarrow \frac{du}{3} = -\sin(3x) dx \end{aligned}$$

$$u_1 = \frac{1}{18} \int \frac{1}{u} du = \frac{1}{18} \ln |u|$$

$$\Rightarrow u_1 = \frac{1}{18} \ln |\cos(3x)|$$

$$u_2 = \int \frac{w_2}{w} dx = \int \frac{1}{6} dx = \frac{x}{6}$$

$$y_p = u_1 y_1 + u_2 y_2$$

$$y_p = \frac{1}{18} \cos(3x) \ln|\cos(3x)| + \frac{x}{6} \sin(3x)$$

The general solution is

$$y = y_c + y_p$$

$$y = C_1 \cos(3x) + C_2 \sin(3x) + \frac{1}{18} \cos(3x) \ln|\cos(3x)| + \frac{x \sin(3x)}{6}$$
